

HACKDAYS 2.0 — GCET GREATER NOIDA

TerraGuard AI

Multimodal Landslide Intelligence & Safe Evacuation Platform

Theme: Best Use of Google Gemini API | Multimodal Vision + Function Calling + Long-Context Reasoning

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Live App: <https://terranguard-ai-fawn.vercel.app> | GitHub: github.com/SKartik03/terranguard-ai

The Problem: Climate Extremes & Deadly Slope Failures

ACUTE CLIMATE CRISIS

- 65 Million People in India live across 0.42M km² of high-hazard terrain (Western Ghats & Himalayas).
- Cloudburst Intensification: Extreme monsoon downpours rapidly oversaturate slopes, triggering deadly flash debris flows (e.g. 350+ casualties in 2024 Wayanad disaster).
- Lifeline Infrastructure Severed: National highways (NH-66, NH-58) cut off, stranding thousands.

CURRENT SYSTEM GAPS

- Coarse District Alerts: Generic rainfall bulletins cover 500+ km² indiscriminately without micro-topographical slope context, causing severe false alarm fatigue.
- Static Hazard Maps: Existing GSI/ISRO atlases are static and do not adapt to live weather.
- No Evacuation Support: Systems stop at red alerts without giving actionable, terrain-aware escape corridors.

OUR VISION

- Location-First AI Engine: Instant assessment anywhere in India using GPS, interactive GIS map, or search.
- Multimodal Geotechnical AI: Leveraging Google Gemini 1.5 to visually analyze slope cracks and orchestrate telemetry tools.
- Actionable Evacuation: Active pathfinding comparing safe high-ridge crests vs debris-blocked valleys.

Proposed Solution: TerraGuard AI Powered by Google Gemini

1. Multimodal Vision Inspection

Gemini 1.5 Vision processes drone surveys, satellite imagery, and on-ground citizen photos to identify tension cracks, crown scarps, and structural retaining wall failures.

2. Agentic Tool Use & Function Calling

Gemini acts as an autonomous disaster reasoning agent, dynamically executing tools like `calculate_dem_slope()`, `query_historical_landslides()`, and `compute_evacuation_corridors()`.

3. Long-Context Historical Grounding

Gemini ingests 165 verified historical landslide archives (ISRO/GSI benchmarks) within its massive context window for zero-hallucination analog matching.

4. Polyglot Civil Defense Dispatches

Generates hyper-local early warning broadcasts in regional languages (Hindi, Malayalam, Bengali, Pahari) tailored for grassroots residents and NDRF commanders.

How Google Gemini API Powers the Core Decision Matrix

Step 1: Input Ingestion

- User GPS / Map Selection
- Live Open-Meteo Weather
- Drone / Camera Slope Photos
- NASA SRTM DEM Gradients

Step 2: Gemini 1.5 Flash/Pro

- Multimodal Image Analysis
- Function Calling & Orchestration
- Geotechnical Safety Factor
- Historical Context Cross-check

Step 3: Analytical Engine

- Layer 1: Rule Safety Net
- Dynamic Weight Renormalization
- Layer 2: Random Forest ML
- Elevation Gradient Derivation

Step 4: Actionable Output

- 0-100 Gauge Risk Score
- Safe High-Ridge Evacuation Path
- Factor-by-Factor Explainability
- Multilingual SMS/Radio Dispatch

The Science: Mohr-Coulomb Dynamics & Dual-Layer Risk Engine

GEOTECHNICAL PRINCIPLES & MATH

- Mohr-Coulomb Failure Dynamics: Shear strength $\tau_f = c' + (\sigma - u) \cdot \tan(\phi')$. Intense rainfall elevates pore pressure (u), diminishing effective stress until catastrophic slope failure occurs.
- Dynamic Weight Renormalization:
 $w_i' = w_i / \sum(w_{avail}) \times 100\%$
When cloud-cover blocks optical satellites (e.g. Sentinel-2 NDVI), remaining factor weights dynamically scale to 100%, preventing artificial score deflation.
- Real-Time DEM Slope Gradients:
Computes $\arctan(\Delta z / \Delta d)$ per coordinate dynamically without heavy GIS servers.

DUAL-LAYER RISK FUSION

- Deterministic Layer 1 Safety Net:
Rigid geotechnical safety rules calculate a normalized physics baseline with zero drift and 100% auditability.
- Empirical Layer 2 Machine Learning:
120-tree Random Forest Classifier trained on 165 verified historical landslide incidents (2012–2026).
- Benchmark Performance:
 - 100% Accuracy & 1.00 ROC-AUC on holdout test calibration (n=33)
 - Feature Importances: Rainfall (34.97%), Soil Moisture (26.24%), Vegetation (20.77%), Land Cover (10.28%), Slope (6.16%).

Beyond Prediction: Dynamic Safe Evacuation Pathfinding

ROUTE A: HIGH RIDGE CREST BYPASS (RECOMMENDED)

- Hazard Score: 18% (LOW HAZARD)
- Topographical Path: Trajectory follows stable granitic ridge spurs elevated well above alluvial drainage channels.
- Clearance: 100% unimpeded clearance to designated community shelter.
- Dynamic Origin Adaptation: Calculates customized hazard-avoidance routes from any user map-click coordinate.

ROUTE B: VALLEY ROADWAY (BLOCKED / IMPASSABLE)

- Hazard Score: 94% (CRITICAL HAZARD)
- Topographical Path: Traverses active monsoonal torrent ravine with severe debris-damming risk.
- Status: BLOCKED at km 2.4 choke-point.
- Gemini Integration: Gemini generates instantaneous police dispatch directives and road closure advisories to divert incoming civilian traffic.

Full-Stack Implementation & Live Cloud Deployment

Production Frontend (Vercel)

- React 18, Vite 6, Custom Dark Glassmorphism
- Leaflet GIS & OpenStreetMap interactive mapping
- Chart.js multi-factor sensitivity radar & curves
- Live URL: <https://terranguard-ai-fawn.vercel.app>

Production Backend (Render)

- Python FastAPI ASGI service on Linux cloud
- SQLite 3 WAL Mode with 165 historical records
- Asynchronous background telemetry scheduler (180s)
- Live API: <https://terranguard-ai-ew30.onrender.com>

14 Geospatial Integrations

- NASA SRTM 30m DEM & ISRO Bhuvan Cartosat
- IMD Pune Gridded Precipitation & Open-Meteo Live
- ESA Sentinel-2 NDVI & NASA SMAP Soil Moisture
- GSI Bhukosh & ISRO Landslide Atlas benchmarks

Zero-Cloud Local Resilience

- Self-contained offline fallback capability
- Runs locally via `start_backend.bat` & `start_frontend.bat`
- 17 automated Pytest suites passing (100% verification)

Multilingual Civil Broadcasts & Transparent Explainability

GEMINI MULTILINGUAL DISPATCHES

- Breaking Language Barriers: Gemini translates and synthesizes technical geotechnical scores into culturally accessible emergency alerts across Indian vernaculars:
 - Hindi: 'चेतावनी: भारी बारिश के कारण व्याथिरी घाट पर भूस्खलन का उच्च जोखिम...'
 - Malayalam: 'മുന്നറിയിപ്പ്: കനത്ത മഴയെത്തുടർന്ന് മേപ്പാടി മേഖലയിൽ മണ്ണിടിച്ചിൽ സാധ്യത...'
 - Bengali & Pahari regional dialects.
- Multi-Tier Early Warning: Automatically formats actionable SMS, WhatsApp dispatches, and community loudspeaker broadcast scripts.

EXPLAINABILITY vs BLACK-BOX

- Factor-by-Factor Audit Trail: Shows exact score points contributed by rainfall, slope angle, regolith saturation, and bedrock condition.
- Deterministic Reproducibility: Re-evaluating identical conditions guarantees identical scores to one decimal place (e.g. 58.1/100 Moderate).
- Gemini Technical Auditor: Officials can converse directly with Gemini to ask: 'Why is Route B flagged critical?' and receive instant physics-backed explanations.

Offline Hackathon Execution Plan at GCET (26 Sep 2026)

Phase 1: Online Screening (Current)

- Full-stack production app deployed (Vercel + Render)
- 165 historical landslide incidents calibrated
- Dynamic slope derivation & routing live

Phase 2: Offline Day - Gemini Live Drone Feed

- Connect Gemini 1.5 Pro multimodal vision to live drone/video camera stream
- Real-time automated crack detection & scarpline polygon extraction

Phase 3: Offline Day - Voice Interaction & Audio

- Gemini voice conversational agent for emergency command centers
- Voice-activated situational inquiries ('Check Wayanad ridge status')

Phase 4: Offline Judging Demonstration

- Table-to-table live scenario simulation (Wayanad monsoon vs Kopergaon)
- Strict deterministic verification + live mobile responsiveness

Why TerraGuard AI Exemplifies 'Best Use of Gemini API'

1. Genuine Multimodal Impact

Not another text wrapper or simple chatbot—leverages Gemini 1.5 Vision, Function Calling, and Long Context to solve a critical, life-threatening disaster problem in India.

2. Real Working Full-Stack Architecture

Already deployed and tested live on Vercel and Render with zero-cloud local fallback and 17 passing automated test suites.

3. Geotechnical Rigor & Explainability

Combines deterministic Mohr-Coulomb slope physics with Random Forest ML, validated on 165 real historical Indian landslides.

4. Ready for Offline Grand Finale

Clear execution roadmap for 26 September at GCET with live drone feed integration and conversational Gemini voice commander.